

German Automotive Sustainability Dashboard

BMW Group vs. Volkswagen Group — CSRD & KPI Analysis (FY 2025)

A comparative sustainability analysis of BMW Group and Volkswagen Group using real audited data extracted directly from their 2025 CSRD/ESRS sustainability reports. The project combines GHG emissions analysis, sustainability KPI benchmarking, and ESRS topic coverage into a single Python dashboard.

What is CSRD?

The **Corporate Sustainability Reporting Directive (CSRD)** is an EU law that requires large companies to disclose sustainability data following the **European Sustainability Reporting Standards (ESRS)**. Both BMW and Volkswagen are subject to CSRD from 2024 onward. Their reports are independently audited — making the data reliable for analysis.

Data Sources

Field	BMW Group	Volkswagen Group
Report	BMW Group Report 2025	VW ESRS Sustainability Report 2025
Page reference	p. 128–129 (GHG), KPI sections	p. 273 (Scope 3), KPI tables
GHG standard	GHG Protocol Scope 3 Standard	GHG Protocol Scope 3 Standard
CSRD standard	ESRS Set 1 (2023)	ESRS Set 1 (2023)
Auditor	PricewaterhouseCoopers GmbH	Ernst & Young GmbH
Audit level	Reasonable assurance	Reasonable assurance

Key Findings

Metric	BMW 2025	VW 2025	Ratio
Scope 1+2 (Mt CO ₂ e)	0.811	2.80	VW 3.5x higher
Scope 3 Total (Mt CO ₂ e)	127.54	883.74	VW 6.9x higher
Total Energy (million MWh)	6.18	19.2	VW 3.1x higher

Metric	BMW 2025	VW 2025	Ratio
Renewable Energy Share	49.4%	40.6%	BMW leads
Water Withdrawal (million m³)	5.70	19.9	VW 3.5x higher
Total Waste (thousand tonnes)	851.8	2,547	VW 3.0x higher
Employees	154,540	602,659	VW 3.9x larger
Scope 3 per employee (t CO ₂ e)	825	1,466	BMW 1.8x more efficient

Key insight: For both companies, 70–85% of all Scope 3 emissions come from the use phase of sold vehicles (Cat 11) — confirming that EV transition is the single most important climate lever for the automotive sector.

What the Script Produces

Running `sustainability_dashboard.py` generates 4 figures and one Excel workbook saved to `output/`:

- **Figure 1** — GHG emissions comparison: Scope 1+2, Scope 3 and total for both companies (2024 vs 2025)
- **Figure 2** — Scope 3 category breakdown: donut charts showing which categories dominate for each company
- **Figure 3** — KPI scorecard: side-by-side comparison of energy, renewables, water, waste and per-employee emissions
- **Figure 4** — CSRD/ESRS topic coverage: which ESRS topics each company reports on and considers material
- **Excel workbook** — full data tables with 3 sheets: GHG emissions, KPI scorecard, data sources

CSRD / ESRS Coverage

ESRS Topic	BMW Reports	BMW Material	VW Reports	VW Material
E1 — Climate Change	✓	✓	✓	✓
E2 — Pollution	✓	✓	✓	✓
E3 — Water & Marine	✓	✗	✓	✓
E4 — Biodiversity	✓	✗	✓	✓

ESRS Topic	BMW Reports	BMW Material	VW Reports	VW Material
E5 — Resource Use	✓	✓	✓	✓
S1 — Own Workforce	✓	✓	✓	✓
S2 — Value Chain Workers	✓	✓	✓	✓
S3 — Communities	✗	✗	✓	✓
S4 — Consumers	✗	✗	✓	✗
G1 — Business Conduct	✓	✓	✓	✓

How to Run

```
bash

pip install pandas matplotlib openpyxl numpy

# Open in Spyder and press F5
# or from terminal:
python sustainability_dashboard.py
```

Project Structure

```
bmw-vw-sustainability-dashboard/
├── sustainability_dashboard.py
├── requirements.txt
├── README.md
└── .gitignore
```

Methodology

Item	Detail
GHG standard	GHG Protocol Corporate Value Chain (Scope 3) Standard (2011)
CSRD standard	ESRS Set 1 — European Sustainability Reporting Standards (2023)
Scope 2 method	Market-based

Item	Detail
Climate science	IPCC AR6 Global Warming Potential (GWP100)
Boundary	Both companies: global operations

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